Shadow Twin Transformer

Exhaustive Mathematical Specification

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1. Purpose

This document rewrites the Transformer and the proposed Shadow Twin extension in a more explicitly mathematical structure, with equations presented as formal blocks rather than loose inline descriptions. The goal is to make the architecture readable as a mathematical system.

2. Notation

Let

$$\begin{aligned} T &:= \text{sequence length}, \\ V &:= \text{vocabulary size}, \\ d &:= d_{\text{model}} \text{ (model width)}, \\ H &:= \text{number of attention heads}, \\ d_k &:= \text{query/key head dimension}, \\ d_v &:= \text{value head dimension}, \\ L &:= \text{number of layers}. \end{aligned}$$

Usually,

$$d = Hd_k, \qquad d_v = d_k.$$

The input token sequence is

$$x = (x_1, x_2, \ldots, x_T), \qquad x_t \in \{1, \ldots, V\}.$$

3. Standard Transformer Mathematics

3.1 Token Embedding

Let the token embedding matrix be

$$E \in \mathbb{R}^{V \times d}.$$

Then each token is embedded as

$$e_t = E[x_t] \in \mathbb{R}^d,$$

and the sequence embedding matrix is

$$X_{\text{emb}} \in \mathbb{R}^{T \times d}.$$

3.2 Positional Information

In additive form,

$$X^{(0)} = X_{\text{emb}} + P,$$

where

$$P \in \mathbb{R}^{T \times d}.$$

For sinusoidal encoding,

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d}}\right),$$
$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d}}\right).$$

3.3 Single-Head Self-Attention

For an input matrix

$$X \in \mathbb{R}^{T \times d},$$

define learned projections

$$W_Q \in \mathbb{R}^{d \times d_k}, \qquad W_K \in \mathbb{R}^{d \times d_k}, \qquad W_V \in \mathbb{R}^{d \times d_v}.$$

Then

$$Q = XW_Q,$$
$$K = XW_K,$$
$$V = XW_V.$$

Raw attention scores are

$$S = QK^\top \in \mathbb{R}^{T \times T},$$

with elementwise form

$$S_{ij} = q_i^\top k_j.$$

Scaled scores are

$$\widetilde{S} = \frac{QK^\top}{\sqrt{d_k}}.$$

With mask $M \in \mathbb{R}^{T \times T}$,

$$A_{\text{scores}} = \frac{QK^\top}{\sqrt{d_k}} + M.$$

Attention weights are

$$A = \text{softmax}(A_{\text{scores}}),$$

where the softmax is row-wise, that is,

$$A_{ij} = \frac{\exp((A_{\text{scores}})_{ij})}{\sum_{m=1}^{T} \exp((A_{\text{scores}})_{im})}.$$

The output is

$$\text{Attn}(Q, K, V) = AV,$$

so for each position i,

$$o_i = \sum_{j=1}^{T} A_{ij} v_j.$$

3.4 Multi-Head Attention

For head $h \in \{1, \ldots, H\}$,

$$\begin{aligned} Q_h &= XW_Q^{(h)}, \\ K_h &= XW_K^{(h)}, \\ V_h &= XW_V^{(h)}. \end{aligned}$$

Then

$$\text{head}_h = \text{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_k}} + M\right) V_h.$$

Concatenate all heads and project:

$$\text{MHA}(X) = \text{Concat}(\text{head}_1, \ldots, \text{head}_H) W_O,$$

where

$$W_O \in \mathbb{R}^{Hd_v \times d}.$$

3.5 Feed-Forward Network

For a token vector $x \in \mathbb{R}^d$,

$$\text{FFN}(x) = W_2\, \phi(W_1 x + b_1) + b_2,$$

where

$$W_1 \in \mathbb{R}^{d_{ff} \times d}, \qquad W_2 \in \mathbb{R}^{d \times d_{ff}}.$$

A common choice is GELU:

$$\text{GELU}(z) = z\Phi(z)$$

with approximation

$$\text{GELU}(z) \approx \frac{1}{2} z \left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}\left(z + 0.044715 z^3\right)\right)\right).$$

A modern alternative is SwiGLU:

$$\text{SwiGLU}(x) = \text{SiLU}(xW_a) \odot (xW_b),$$

followed by

$$\text{FFN}(x) = \text{SwiGLU}(x) W_o.$$

3.6 Layer Normalization

For $x \in \mathbb{R}^d$,

$$\mu = \frac{1}{d}\sum_{i=1}^{d} x_i,$$
$$\sigma^2 = \frac{1}{d}\sum_{i=1}^{d}(x_i - \mu)^2,$$

and

$$\mathrm{LN}(x) = \gamma \odot \frac{x-\mu}{\sqrt{\sigma^2+\varepsilon}} + \beta.$$

3.7 Residual Connection

For a sublayer F,

$$Y = X + F(X).$$

3.8 Transformer Layer Forms

Post-LN form:

$$\widehat{X} = \mathrm{LN}\big(X^{(\ell-1)} + \mathrm{MHA}(X^{(\ell-1)})\big),$$
$$X^{(\ell)} = \mathrm{LN}\big(\widehat{X} + \mathrm{FFN}(\widehat{X})\big).$$

Pre-LN form:

$$\widehat{X} = X^{(\ell-1)} + \mathrm{MHA}(\mathrm{LN}(X^{(\ell-1)})),$$
$$X^{(\ell)} = \widehat{X} + \mathrm{FFN}(\mathrm{LN}(\widehat{X})).$$

3.9 Stacking Layers

For $\ell = 1, \ldots, L$,

$$X^{(L)} = \mathrm{TransformerLayer}_L \circ \cdots \circ \mathrm{TransformerLayer}_1(X^{(0)}).$$

Optionally,

$$H = \mathrm{LN}(X^{(L)}).$$

3.10 Output Head and Probabilities

Let

$$W_{\mathrm{out}} \in \mathbb{R}^{d\times V}.$$

Then

$$Z = HW_{\text{out}} + b_{\text{out}}, \qquad Z \in \mathbb{R}^{T\times V}.$$

At position t,

$$p_t = \text{softmax}(z_t),$$

and for vocabulary symbol v,

$$p(y_t = v \mid x_{\le t}) = \frac{e^{z_{t,v}}}{\sum_{u=1}^{V} e^{z_{t,u}}}.$$

Often,

$$W_{\text{out}} = E^{\top}.$$

3.11 Autoregressive Training Objective

The factorization is

$$P(x_1, \dots, x_T) = \prod_{t=1}^{T} P(x_t \mid x_{<t}).$$

The negative log-likelihood loss is

$$\mathcal{L}_{\text{LM}} = -\sum_{t=1}^{T} \log P(x_t \mid x_{<t}).$$

Equivalent tokenwise cross-entropy form:

$$\mathcal{L}_{\text{LM}} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{v=1}^{V} y_{t,v} \log p_{t,v}.$$

4. Shadow Twin Extension

4.1 Two Coupled Streams

The architecture has two evolving states:

$$\begin{aligned} X^{(\ell)} &\in \mathbb{R}^{T\times d} && \text{text stream,} \\ S^{(\ell)} &\in \mathbb{R}^{M\times d} && \text{shadow stream.} \end{aligned}$$

The text stream processes prompt tokens. The shadow stream processes retrieved world-memory items.

4.2 External World Memory

Let the external database be

$$\mathcal{D} = \{m_1, m_2, \dots, m_N\}.$$

Each item has embedding

$$u_i \in \mathbb{R}^d,$$

so collectively

$$U \in \mathbb{R}^{N \times d}.$$

4.3 Initialization

The text branch starts as usual:

$$X^{(0)} = E[x] + P.$$

Define a global text query

$$q_{\text{global}}^{(0)} = \text{Pool}(X^{(0)}) W_q^{\text{mem}}.$$

Memory scores are

$$r_i^{(0)} = \frac{q_{\text{global}}^{(0)} \cdot u_i}{\sqrt{d}}.$$

Take the top-M records:

$$\mathcal{I}^{(0)} = \text{TopM}(r_1^{(0)}, \ldots, r_N^{(0)}).$$

Then

$$S^{(0)} = U[\mathcal{I}^{(0)}] \in \mathbb{R}^{M \times d}.$$

4.4 Text Stream Update

The text stream first performs self-attention:

$$\widetilde{X}^{(\ell)} = X^{(\ell-1)} + \text{MHA}_{xx}^{(\ell)}(\text{LN}(X^{(\ell-1)})).$$

Then it attends to the shadow stream:

$$\overline{X}^{(\ell)} = \widetilde{X}^{(\ell)} + \text{MHA}_{xs}^{(\ell)}(Q = \text{LN}(\widetilde{X}^{(\ell)}),\, K = \text{LN}(S^{(\ell-1)}),\, V = \text{LN}(S^{(\ell-1)})).$$

Expanded head form:

$$\text{MHA}_{xs}(X, S) = \text{Concat}_{h=1}^{H} \left[\text{softmax}\left(\frac{(XW_Q^{x,h})(SW_K^{s,h})^\top}{\sqrt{d_k}} \right) (SW_V^{s,h}) \right] W_O^{xs}.$$

Then the feed-forward update is

$$X^{(\ell)} = \overline{X}^{(\ell)} + \text{FFN}_x^{(\ell)}(\text{LN}(\overline{X}^{(\ell)})).$$

4.5 Shadow Stream Update

The shadow state reasons over retrieved memories via self-attention:

$$\widetilde{S}^{(\ell)} = S^{(\ell-1)} + \mathrm{MHA}_{ss}^{(\ell)}(\mathrm{LN}(S^{(\ell-1)})).$$

Then it attends to the text stream:

$$\overline{S}^{(\ell)} = \widetilde{S}^{(\ell)} + \mathrm{MHA}_{sx}^{(\ell)}(Q = \mathrm{LN}(\widetilde{S}^{(\ell)}),\ K = \mathrm{LN}(X^{(\ell-1)}),\ V = \mathrm{LN}(X^{(\ell-1)})).$$

Expanded head form:

$$\mathrm{MHA}_{sx}(S, X) = \mathrm{Concat}_{h=1}^{H}\left[\mathrm{softmax}\left(\frac{(SW_Q^{s,h})(XW_K^{x,h})^\top}{\sqrt{d_k}}\right)(XW_V^{x,h})\right]W_O^{sx}.$$

4.6 Shadow Re-Retrieval

The shadow twin can re-query the full database at every layer. Define

$$q_{\text{shadow}}^{(\ell)} = \mathrm{Pool}(\overline{S}^{(\ell)})W_q^{\text{shadow}} + \mathrm{Pool}(X^{(\ell)})W_q^{\text{text2mem}}.$$

Then

$$r_i^{(\ell)} = \frac{q_{\text{shadow}}^{(\ell)} \cdot u_i}{\sqrt{d}}.$$

A more general metadata-aware score is

$$r_i^{(\ell)} = g(q_{\text{shadow}}^{(\ell)}, u_i, \mathrm{meta}_i).$$

Take the new top-M records:

$$\mathcal{I}^{(\ell)} = \mathrm{TopM}(r_1^{(\ell)}, \ldots, r_N^{(\ell)}),$$

and define the refreshed retrieval state

$$R^{(\ell)} = U[\mathcal{I}^{(\ell)}].$$

4.7 Fusion of Old and New Shadow Memory

A simple merged form is

$$\widehat{S}^{(\ell)} = W_m \begin{bmatrix} \overline{S}^{(\ell)} \\ R^{(\ell)} \end{bmatrix}.$$

A gated fusion alternative is

$$\begin{aligned} G^{(\ell)} &= \sigma(\overline{S}^{(\ell)}W_g^{(1)} + R^{(\ell)}W_g^{(2)}), \\ \widehat{S}^{(\ell)} &= G^{(\ell)} \odot \overline{S}^{(\ell)} + (1 - G^{(\ell)}) \odot R^{(\ell)}. \end{aligned}$$

Then the shadow feed-forward update is

$$S^{(\ell)} = \widehat{S}^{(\ell)} + \mathrm{FFN}_s^{(\ell)}(\mathrm{LN}(\widehat{S}^{(\ell)})).$$

4.8 Mutual Informing Messages

The shadow stream can send a distilled message to the text stream:

$$m_{s\to x}^{(\ell)} = \mathrm{Pool}(S^{(\ell)})W_{s\to x}.$$

Broadcast across token positions:

$$M_{s\to x}^{(\ell)} = \mathbf{1}_T (m_{s\to x}^{(\ell)})^\top.$$

Then inject:

$$X^{(\ell)} \leftarrow X^{(\ell)} + M_{s\to x}^{(\ell)}.$$

Likewise the text stream sends an intent summary:

$$m_{x\to s}^{(\ell)} = \mathrm{Pool}(X^{(\ell)})W_{x\to s}.$$

Broadcast across memory items:

$$M_{x\to s}^{(\ell)} = \mathbf{1}_M (m_{x\to s}^{(\ell)})^\top.$$

Then

$$S^{(\ell)} \leftarrow S^{(\ell)} + M_{x\to s}^{(\ell)}.$$

4.9 Compact Coupled Layer Form

The coupled system can be summarized as

$$\begin{aligned} X^{(\ell)} &= X^{(\ell-1)} + A_{xx}^{(\ell)} + A_{xs}^{(\ell)} + C_{s\to x}^{(\ell)} + F_x^{(\ell)}, \\ S^{(\ell)} &= S^{(\ell-1)} + A_{ss}^{(\ell)} + A_{sx}^{(\ell)} + R^{(\ell)} + C_{x\to s}^{(\ell)} + F_s^{(\ell)}. \end{aligned}$$

4.10 Full Layer-by-Layer Mathematical Form

Initialization:

$$\begin{aligned} X^{(0)} &= E[x] + P, \\ S^{(0)} &= \mathrm{Retrieve}(\mathcal{D}, \mathrm{Pool}(X^{(0)})). \end{aligned}$$

For each layer $\ell = 1, \ldots, L$:

$$
\begin{aligned}
\widetilde{X}^{(\ell)} &= X^{(\ell-1)} + \mathrm{MHA}_{xx}^{(\ell)}(\mathrm{LN}(X^{(\ell-1)})), \\
\overline{X}^{(\ell)} &= \widetilde{X}^{(\ell)} + \mathrm{MHA}_{xs}^{(\ell)}(\mathrm{LN}(\widetilde{X}^{(\ell)}), \mathrm{LN}(S^{(\ell-1)})), \\
X^{(\ell)} &= \overline{X}^{(\ell)} + \mathrm{FFN}_{x}^{(\ell)}(\mathrm{LN}(\overline{X}^{(\ell)})), \\
\widetilde{S}^{(\ell)} &= S^{(\ell-1)} + \mathrm{MHA}_{ss}^{(\ell)}(\mathrm{LN}(S^{(\ell-1)})), \\
\overline{S}^{(\ell)} &= \widetilde{S}^{(\ell)} + \mathrm{MHA}_{sx}^{(\ell)}(\mathrm{LN}(\widetilde{S}^{(\ell)}), \mathrm{LN}(X^{(\ell-1)})), \\
R^{(\ell)} &= \mathrm{Retrieve}(\mathcal{D}, \mathrm{Pool}(X^{(\ell)}), \mathrm{Pool}(\overline{S}^{(\ell)})), \\
\widehat{S}^{(\ell)} &= \mathrm{Fuse}(\overline{S}^{(\ell)}, R^{(\ell)}), \\
S^{(\ell)} &= \widehat{S}^{(\ell)} + \mathrm{FFN}_{s}^{(\ell)}(\mathrm{LN}(\widehat{S}^{(\ell)})), \\
X^{(\ell)} &\leftarrow X^{(\ell)} + \mathrm{Broadcast}(\mathrm{Pool}(S^{(\ell)})W_{s\to x}), \\
S^{(\ell)} &\leftarrow S^{(\ell)} + \mathrm{Broadcast}(\mathrm{Pool}(X^{(\ell)})W_{x\to s}).
\end{aligned}
$$

5. Output of a Shadow Twin LLM

5.1 Standard Versus Shadow-Twin Prediction

A standard LLM predicts

$$P(y_t \mid y_{<t}, x).$$

A Shadow Twin model predicts

$$P(y_t \mid y_{<t}, x, \mathcal{D}, S_t).$$

That is, the output depends not only on prompt text and previous generated tokens, but also on the retrieved world-memory state.

5.2 Fused Generation State

A mathematical generation pipeline can be written as

$$
\begin{aligned}
h_t^{\mathrm{text}} &= \mathrm{TextTransformer}(x, y_{<t}), \\
h_t^{\mathrm{shadow}} &= \mathrm{ShadowRetriever}(\mathcal{D}, h_t^{\mathrm{text}}), \\
h_t^{\mathrm{fused}} &= \mathrm{Fuse}(h_t^{\mathrm{text}}, h_t^{\mathrm{shadow}}), \\
z_t &= W_{\mathrm{out}} h_t^{\mathrm{fused}}, \\
p_t &= \mathrm{softmax}(z_t).
\end{aligned}
$$

5.3 Final Hidden-State Fusion

One token-level fusion form is

$$h_t^* = W_f \begin{bmatrix} x_t^{(L)} \\ c_t^{(L)} \end{bmatrix},$$

where $c_t^{(L)}$ is a knowledge-context vector obtained from the shadow stream. Then

$$z_t = h_t^* W_{\text{out}}.$$

Hence

$$p(y_t \mid x_{\leq t}, \mathcal{D}) = \text{softmax}(z_t).$$

5.4 Confidence-Aware Output Policy

If retrieval scores are r_i, then low confidence can be detected by

$$\max_i r_i < \tau,$$

for some threshold τ. In that case the model may switch to a cautious answer policy. This formalizes the possibility that the model says it is uncertain rather than fabricating a fluent but unsupported answer.

5.5 Time-Aware Memory Records

A more realistic world-memory item can be written as

$$(u_i, c_i, t_i, s_i),$$

where

$$\begin{aligned} u_i &: \text{memory embedding}, \\ c_i &: \text{confidence}, \\ t_i &: \text{timestamp}, \\ s_i &: \text{source metadata}. \end{aligned}$$

A retrieval score can then be time-aware and source-aware:

$$r_i^{(\ell)} = \alpha \ \text{sim}(q^{(\ell)}, u_i) + \beta \, c_i + \gamma \ \text{TimeScore}(t_i) + \delta \ \text{SourceScore}(s_i).$$

6. Training Objectives for the Shadow Twin Model

The main language-model loss remains

$$\mathcal{L}_{\text{LM}} = -\sum_{t=1}^{T} \log P(x_t \mid x_{<t}, \mathcal{D}).$$

A retrieval relevance loss can be written as

$$\mathcal{L}_{\text{ret}} = -\log \frac{\exp(\text{sim}(q, u^+))}{\exp(\text{sim}(q, u^+)) + \sum_{u^-} \exp(\text{sim}(q, u^-))}.$$

A semantic consistency term between streams can be

$$\mathcal{L}_{\text{cons}} = \left\| \text{Pool}(X^{(L)}) - \text{Pool}(S^{(L)}) \right\|^2 .$$

A faithfulness loss can be written abstractly as

$$\mathcal{L}_{\text{faith}} = \text{Divergence}(\text{generated claim}, \text{retrieved support}).$$

The combined objective is

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_1 \mathcal{L}_{\text{ret}} + \lambda_2 \mathcal{L}_{\text{cons}} + \lambda_3 \mathcal{L}_{\text{faith}}.$$

7. Compact Final System Expression

The entire Shadow Twin language model can be summarized as

$$\text{Output} = f\big(\underbrace{\text{Prompt Text}}_{\text{text stream}}, \underbrace{\text{Retrieved World Knowledge}}_{\text{shadow stream}}, \underbrace{\text{Cross Stream Communication}}_{\text{mutual informing}}\big).$$

A compact probability statement is

$$\boxed{P(y_t \mid y_{<t}, x, \mathcal{D}, S_t)}.$$

A compact architectural identity is

$$\boxed{\text{Shadow Twin LLM} = \text{Language Model} + \text{Knowledge System} + \text{Verification Loop}.}$$

8. Practical Note

The phrase "knowledge of everything that ever happened" is mathematically useful as an idealized target but not physically exact. Any real database $\mathcal{D}$ would be incomplete, noisy, contradictory, time-dependent, and uncertain. The equations above therefore describe the architecture in ideal form, while the metadata-aware and confidence-aware terms make it more realistic.